

Review

Automated diagnosis and classification of liver cancers using deep learning techniques: a systematic review

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Abstract

Liver cancer is one of the main causes of cancer-related mortality globally. It is a rising threat with over 700,000 deaths and 800,000 new cases annually. To tackle this, deep learning (DL) artificial intelligence (AI) tools have been researched extensively to create predictive models that aid in the early diagnosis and classification of liver cancers. The liver, however, is prone to metastasis from various gastrointestinal cancers, making it difficult to correctly classify tumors present in the organ. This paper is a thorough systematic evaluation of several published studies that used deep learning algorithms to diagnose, classify, and predict common liver tumors, including but not limited to Hepatocellular carcinoma and Intrahepatic cholangiocarcinoma. Articles selected in this review have been published between 2016 and 2024 from Google Scholar, Research Gate, Science Direct, IEEE Xplore, Springer Link, and other resources as per the PRISMA guidelines. Amongst these, the highest recorded accuracy was 99.38%, with other studies showing close results at 97% and below. The highest recorded sensitivity was 100%. Selected studies were also analyzed based on their choice of dataset, DL algorithm, and preprocessing techniques. Magnetic resonance imaging and computed tomography (CT) are by far the preferred medium for imaging data because of their high resolution; however, CT is used more often because it's cheaper and provides a better view of extrahepatic space. Genetic datasets showed great potential in differentiating between primary and secondary liver cancers. Nonetheless, several gaps were identified across the current literature. The problem of accurately differentiating cancer types in the liver still persists. However, the use of 3D imaging datasets aids in this significantly. Though, it is clear that 3D datasets are underutilized. Dynamic imaging is also beneficial and should be considered more often. For future research, we strongly recommend the use of Gaussian Mixture Model, Watershed Transform, Sparsity classification, DLIR and Siamese algorithms as they each showed promising results. Lastly, we have shown a benchmark framework for a predictive AI model that can be referred to develop future models. Though numerous past methods have produced excellent prediction results, the prevalence of liver cancer-related deaths has not yet been reduced. To address the difficulties in the field of cancer prediction, more thorough research is required, along with the development of more efficient decision support systems using deep learning.

Article Highlights

- The article provides a comprehensive evaluation of various diagnostic approaches used for liver cancers and their potential in Deep Learning diagnostic models.
- We present a benchmark framework for an AI-based computational model for the diagnosis of diseases
- The article presents novel approaches for preprocessing and diagnostic techniques for liver cancer diagnosis

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Keywords Liver cancer · Artificial intelligence · AI-based prediction · Diagnosis · Decision support system · Deep learning

1 Introduction

Cancer is a genetic disease caused by changes in the genes that control cell function, especially cell growth and division [1]. It can be inherited, caused by damage to DNA, or simply mutations and errors occurring during cell division processes. Cancer formation is described as uncontrolled cell division, where the body's cells continue to proliferate, ignoring signals, such as cyclins, to manage the rate [2]. All tumors and patients have a different combination of genetic factors and mutations influencing the growth of their tumor, making cancer difficult to fight [2, 3]; hence, proactive measures such as early diagnosis and prediction are vital and can save numerous lives. As of 2022, WHO claimed there were 20 million new cases of cancer worldwide [4]. The diagnosis and classification of various types of cancer become difficult because of metastasis and the presence of tissue changes that are not cancer. Metastasis is the process by which cancer cells break away from the original (primary) tumor and spread via blood, lymphatic system, etc., to new locations in the body where they form new (secondary) tumors [5]. Additionally, Hyperplasia (Hypergenesis) is another process that can make cancer diagnosis challenging. Hyperplasia occurs when cells in a tissue multiply rapidly, causing extra cells to build up [6]. While Hyperplasia may be a precursor to cancer or other abnormalities, it can also be a build-up of normal cells, making diagnosis difficult and confusing [7]. Similarly, dysplastic cells (abnormal cells), which are usually not cancerous, appear anomalous when observed. This can cause problems when diagnosing cancer through imaging techniques. One example of this is the difficulty that occurs when differentiating between dysplastic nodules and early Hepatocellular Carcinoma in the liver [8].

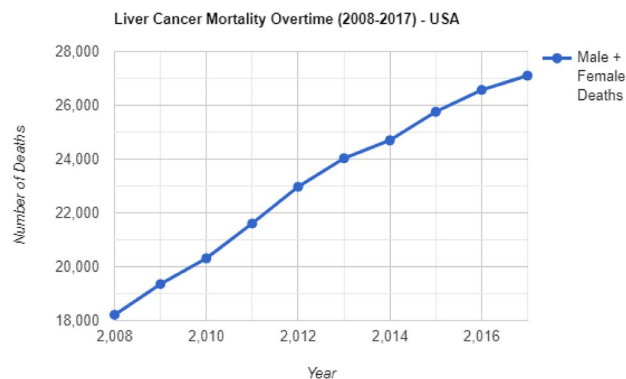
Liver cancer is one of the most prevalent cancer variants, regarded as the 6th most-diagnosed cancer in 2020 [9]. New research shows that more than 800,000 people are diagnosed with liver cancer each year [10]. With this, as of 2020, liver cancer accounted for 8.3% of total cancer deaths [11]. Liver cancer symptoms also present late, meaning patients showing symptoms are likely to have serious cases [12]. The most common forms of primary liver cancers are Hepatocellular Carcinoma (HCC) and Intrahepatic Cholangiocarcinoma (ICC) [13]. However, because of the liver's dual blood supply, it is highly susceptible to secondary tumors (metastasis) from various other gastrointestinal cancers [14]. Furthermore, the sinusoidal epithelium, which comprises cells that make up the walls of the hepatic sinusoids (specialized capillaries), contains fenestrations that enable malignant cells to penetrate the walls and enter the liver parenchyma [15, 16]. Metastases also occur because of the connection between the blood supply of the intestines and the liver via the portal vein. To support this, research showed that around 70% of patients with colorectal cancers eventually develop liver metastases as well [17]. The prevalence of metastasized tumors in the liver makes the classification of liver tumors extremely difficult.

Liver cancer as a threat is also growing. Figure 1 shows the persistent increase in liver cancer death rate in the USA over a decade.

Research published in the Journal of Hepatology predicted that the number of new liver cancer cases annually would increase by around 50% by 2040—meaning 1.4 million new yearly cases. This emphasizes the need for efficient diagnostic systems that can intake large amounts of data, and thus handle more patients more quickly [18].

Alpha-fetoprotein tumor marker tests, liver function tests, and biopsies are all common, yet invasive, diagnosis methods for liver cancers [19]. Imaging, however, is also a viable tool. Research on artificial intelligence-deep learning (AI-DL) models for the automated diagnosis and classification of cancers using imaging like CT or MRI has been rising. There have been great strides in the use of AI for medical purposes, like faster diagnosis, improved accuracies, specificity, and sensitivities comparable to or

Fig. 1 Liver cancer deaths in the USA per year from 2008 to 2017 [95]



outperforming expert radiologists in the respective fields. Thus, AI-based models have significant potential for clinical application for cancer diagnosis. Computer-aided diagnosis provides higher than before accuracy and also enables minimal manual interaction. This also makes these tools applicable for use in areas with few medical professionals. With this, this review aims to highlight the progression of deep learning in the diagnosis and classification of liver cancers. Various novel models are analyzed to understand what is working and what is not. Integrative models have also been analyzed, where additional modifiers have been incorporated to improve feature extraction, liver lesion segmentation, delineation, etc. Both of which are highly useful for tumor classification. The advantages of the datasets and models used are also explored. This paper is a comprehensive review intended to highlight the contribution of different papers in developing techniques for the purpose of tackling liver cancer diagnosis and to understand what can be improved.

1.1 Contributions of manuscript

This study comprises an exhaustive systematic review of various published studies utilizing deep learning algorithms to diagnose, classify and prognose common liver cancers. Also, we provide reasoning behind the complications of diagnosing liver cancers, with an emphasis on metastases. Our primary research contributions are as follows:

- A detailed understanding of liver cancer detection methods utilizing conventional diagnostic techniques is examined in the study. And, this study provides comparisons between popularly used diagnostic procedures for liver cancers. This is done to help decide and approach the dataset to use for future such computational models.
- The study proposes an efficient AI-based Computational framework for the diagnosis of cancer.
- The review highlights various gaps identified among the studies analyzed, identifying the scopes for improvement. The challenges associated with detecting liver cancer through deep learning are emphasized. The future scope of current studies has also been noted.
- Suggestions for future computational models are also presented, backed with evidence extracted from the analysis of existing works.

1.2 Organization of manuscript

Section 2: Here we show the research process used for this review, following the PRISMA guidelines. The criteria used to select the studies are shown, and the research investigations being answered are also listed.

Section 3: This section details the background knowledge, and explains various popular diagnosis methodologies used around the globe. We primarily discuss methods that are used to train predictive AI models. Additionally, this section also details the structure and function of popular AI Deep Learning algorithms that are discussed later. We have also made an attempt to explain the workings of the most commonly used DL algorithms, namely ANN, CNN, FCN, and SVM algorithms.

Section 4: The literature survey section conducts a comparative analysis between diagnostic imaging tools used for training computational predictive models, and elaborates use of 2D and 3D datasets for the same. Lastly, this section provides the literature review, where existing studies on liver cancer predictive models are systematically analyzed based on their dataset used, objective, challenges, and future scope.

Section 5: In this section, we show a benchmark framework for an AI-based predictive model that can be built on to create future models.

Section 6: This section provides solutions to all the research investigations we mentioned earlier. We state the evaluation parameters used for AI models and provide descriptions of popular liver cancer datasets of many types. This section also discusses the research gaps that have been identified through the literature review, and we present detailed, evidence-backed suggestions for future computational models and areas for research.

Section 7: The conclusion section summarizes the future potential of the use of Deep Learning in the field of disease diagnosis, and in terms of liver cancers.

2 Materials and methods

This review has been conducted in accordance with the PRISMA guidelines. The literature review was conducted during the months of January and February 2024, using Google Scholar, Research Gate, Science Direct, IEEE Xplore, etc. as the sources for the works. Each of these sources are search engines filled with academic research only. For the papers gained

from manual searches, keywords such as these were used: “Liver Cancer Diagnosis”, “Deep Learning”, “Machine Learning”, “Liver Cancer Classification”, “Artificial Intelligence” and “HCC and ICC diagnosis”. Overall, original research papers with the objective of using artificial intelligence ML/DL models for the classification, diagnosis, and prognosis of various liver cancers were incorporated in this review. The exhaustive selection criteria followed to attain all the research articles can be seen in Table 1. These criteria were based on the PRISMA flowchart given in Fig. 2.

Table 1 below shows the inclusion and exclusion criteria that were used to select the studies for this review. The PRISMA Flow Diagram has been based on these criteria.

2.1 Research investigations

This study aims to provide solutions for 4 research investigations. Below are the 4 investigations:

- *Research Investigation 1:* What are the evaluation parameters for AI ML/DL predictive models?
- *Research Investigation 2:* List commonly available datasets for liver tumors
- *Research Investigation 3:* What research gaps have been identified in recent studies?
- *Research Investigation 4:* What potential research opportunities can be suggested?

3 Background study

This section provides an overview of various diagnostic techniques used for liver cancer, including the most commonly utilized imaging (MRI, CT and US) and invasive techniques (miRNA and DNA methylation). Also, the section digs into methods hospitals use to maximize the efficacy of these tools through processes like dynamic or contrast-enhanced imaging. We then discuss the most common and efficient Deep Learning algorithms implemented for predictive AI models.

3.1 Diagnostic procedures commonly used for liver cancers

This section provides descriptions of various diagnostic methods used for liver cancers, specifically those that are commonly used as datasets for training Artificial Intelligence models.

3.1.1 Imaging techniques

- *MRI (Magnetic resonance imaging):* MRI is a non-invasive imaging technique used to take detailed images of organs, bones, muscles, and blood vessels. This is done by passing current to coiled wires, to create an electrical field in the patient’s body [20]. Radiowaves are then sent through the body to create a detailed image of internal structures. When time and money are not an issue, MRI is usually the preferred choice over CT when soft tissue or organs need to be viewed. This is because MRI has been shown to be a better indicator of differences between normal and abnormal tissue [21].
- *CT (Computerized tomography):* CT scanning is another commonly used non-invasive screening technique. It is done through the use of X-ray waves to create an image of the body. CT scans use machines that rotate rapidly in circles around the patient so that X-ray beams are shot at many different angles, providing greater detail and making it superior to X-ray scans by themselves [22]. Once scanned, CT data is then cut into cross-sectional 2D images called slices (also done in MRI), and this is what specialists review to make a medical decision. [23] It is commonly used to detect tumors and internal bleeding/damage [22].
- *Ultrasound imaging:* Ultrasound scans (non-invasive) use high-frequency sound waves to create images of internal structures in the body [24]. The scan is performed through an ultrasound transducer. Ultrasound waves are sent into the body and move through it. The waves that reflect back are detected by the transducer, and used to produce an image of the structures inside the body [25]. Liver ultrasound has many benefits; it can show signs of fat storage in your liver (steatotic liver disease), hepatitis, fibrosis or cirrhosis, and liver lesions [26].

Table 1 Inclusion and exclusion parameters

S.no	Parameters	Inclusion standards	Exclusion standards
1	Period	Original academic research conducted between 2016 and 2024	Original academic research conducted before 2016
2	Investigations	Academic research focused on the diagnosis of different liver cancers	Academic research focused on diseases other than liver cancers
3	Comparator	Academic research works aiming to predict the diagnosis, prognosis, or classification of liver cancers	Academic research works proposing predictive models other than for liver cancers
4	Methodology	Academic research articles using ML/DL methods	Academic research articles using methods other than ML/DL
5	Design of Study	Original research works comprising experimental results	Review papers, case studies, or papers in languages other than English

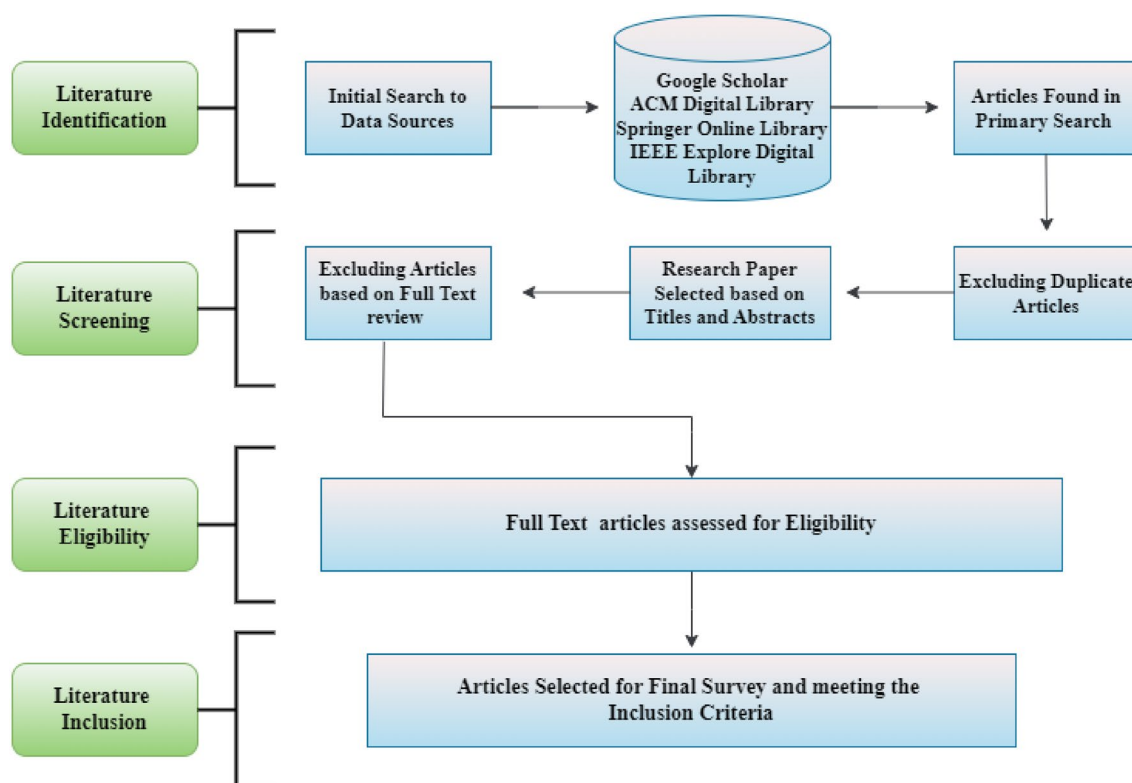


Fig. 2 PRISMA flow diagram

3.1.1.1 Use of contrast agents Contrast agents are commonly used in MRI and CT screenings. A contrast is a substance injected into the body, orally or through an IV [22]. The use of contrast-enhanced MRI (CE-MRI) or contrast-enhanced CT (CECT) is common when diagnosing liver masses because the contrast improves the delineation of the suspected region in the patient, making the reports easier to review. Contrast agents can have side effects on the patient and are therefore used only when necessary. In our context, contrast-enhanced CT is recommended for detecting masses and malignancies [27].

3.1.1.2 Dynamic imaging Dynamic imaging is used with both CT and MRI screenings. Dynamic imaging is performed by taking sequential images (CT or MRI) over a period of time to understand the intricacies of tissue microvasculature. It can be used to capture changes in the body over time [28]. Take CT for example: The functionality of regular CT and dynamic CT is identical, except that dynamic CT involves taking a series of series of images continuously over time. This allows for monitoring of internal processes like blood flow. Differences in blood volume and vascular permeability are evident in dynamic imaging and can be indicative of problems, like cancers [29].

3.1.2 Alternative methods for cancer diagnosis used in AI models

Other than imaging, DNA/blood profiling is a common, but invasive, tool for diagnosis. However, profiling data is also commonly used to train AI predictive models, thus is discussed here.

3.1.2.1 Micro RNA (miRNA) profiling for cancer diagnosis Micro RNA (miRNA) are small non-coding RNA molecules involved in the regulation of gene expression. These molecules account for roughly 1–5% of the human genome [30], and regulate over half of all protein-encoding genes [31]. miRNA regulate gene expression by binding to messenger RNA (mRNA) molecules in the cell. Once bound, the mRNA will either be destroyed or recycled. Such processes reduce the rate of translation and therefore the protein being coded for by that mRNA is underexpressed [32]. It is possible for one miRNA molecule to bind and affect many mRNAs, and multiple miRNAs affect only one. Because of this, a single

miRNA is able to regulate and control various cellular processes, including proliferation, apoptosis, etc. [31]. This and more research on miRNA concluded that numerous human diseases showed an irregular expression of miRNA [33].

Additionally, miRNA that are secreted outside of cells into extracellular fluid in exosomes, known as circulating miRNA, have been identified as biomarkers [34] for various human diseases and even cancers [35]. miRNA profiling has shown that miRNA expression is different in normal and tumor tissues, and can therefore be used to distinguish between them [31]. This discovery brings a lot of potential for early diagnosis of neoplasms, allowing for proactive action and specialized treatment methods. The latest advances in sequencing platforms for whole-genome analysis of miRNA expression have identified aberrantly expressed miRNA profiles corresponding to different cancers (breast, colorectal, and prostate cancers, etc.) [31]. Such profiles are what is used to train Artificial Intelligence models to be able to identify miRNA expression profiles that correspond to a specific cancer.

3.1.2.2 DNA methylation profiling for cancer diagnosis DNA Methylation is an epigenetic tag involving the transfer of a methyl group to the 5-carbon position of a cytosine, usually located in the promotor region of a gene [36]. Methylation of a gene has the ability to regulate its transcription: methylation of cytosine in the promoter region of a gene inhibits the binding of transcription factors to that gene, thus downregulating the rate of transcription. Methylation also acts by recruiting proteins involved in gene expression, slowing down the process [36]. DNA methylation is common in cells for development; however, abnormal DNA methylation commonly disrupts signaling mechanisms between cells, resulting in various diseases including cancer [37]. To understand and profile the pattern of methylation in a cell, CpG sites—regions on DNA with a higher density of cytosine and guanine nucleotides—are analyzed [36]. More specifically, DMRs (differentially methylated regions) are analyzed, which are multiple consecutive CpG sites and occur throughout the genome. As mentioned before, DMRs are found mostly in the promoter regions of genes [38].

In order to utilize DNA methylation as a biomarker for disease, we look for abnormalities such as hypermethylation and hypomethylation. Hypermethylation, an increase in methylation, of the promoter sequence results in gene silencing, whereas hypomethylation is associated with upregulation [38]. For cancers, research has linked the methylation of promoter sequences of tumor suppressor genes to be an early sign of cancer [39]. Often, the development of cancer is caused by the hypermethylation of tumor suppressor genes and hypomethylation of oncogenes [40]. There are various methods to quantify methylation levels, the most popular of which is whole-genome bisulfite sequencing (WGBS) [41]. However, to understand the extent of the regulation mechanism of DNA methylation, the expression profiles of the corresponding mRNA can also be checked [40].

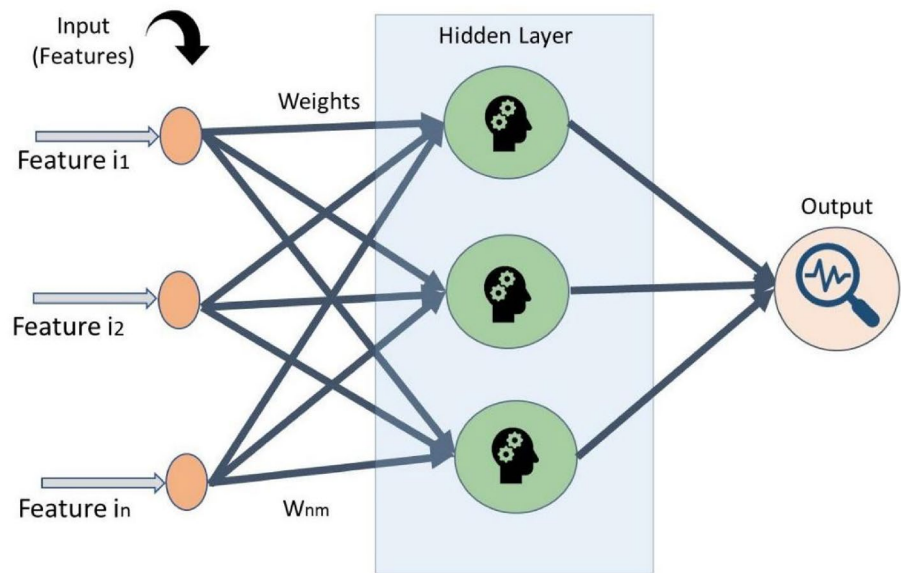
Certain genes such as RASSF1A and p16 have been shown to commonly be highly methylated in many cancers. However, there are also other specific genes that are methylated for specific cancers, and this knowledge can be used to differentiate between them [42]. When training AI to classify cancers or tumors, DNA methylation profiles of specific cancer patients can be used, so that specific genes and methylation sites can be identified that are abnormally methylated and thus probably associated with that cancer. Methylation profiles of patients can also then be inputted into AI to identify abnormalities in methylation sites and expression amount, which can then be cross-checked to existing data on methylation patterns in different cancers.

3.2 Deep learning models

The following are some of the most popularized and efficient Deep Learning algorithms used for predictive AI. The implementation of these algorithms is observed in Sect. 4.

3.2.1 Artificial neural networks (ANNs)

ANNs are computer network systems built of layers of code that mimic the structure and functionality of the human nervous system. The human nervous system and neural network comprise signals being transported across neurons, where sensory neurons send signals through the nervous system, eventually producing an output through axons to effectors. ANN is similar to this, as it is a hierarchical network made up on many neurons, created to learn and synthesize new information from its input information and experiences—similar to the ability of the human brain. Without any instructions, ANNs are able to learn from examples that they are trained on, and thus able to perform the desired task. Their efficiency and performance depend directly on the amount of data they have been trained on. ANNs are built up of 3 layers: input layer, hidden layer, and output layer. The number of hidden layers can vary as the architecture and requirements of the models change [43, 44] (Fig. 3).

Fig. 3 Architecture of ANN

In the structure of an ANN, interconnected nodes called neurons are organized in layers. Input data is processed through these layers of neurons, which adjust weights and identify patterns.

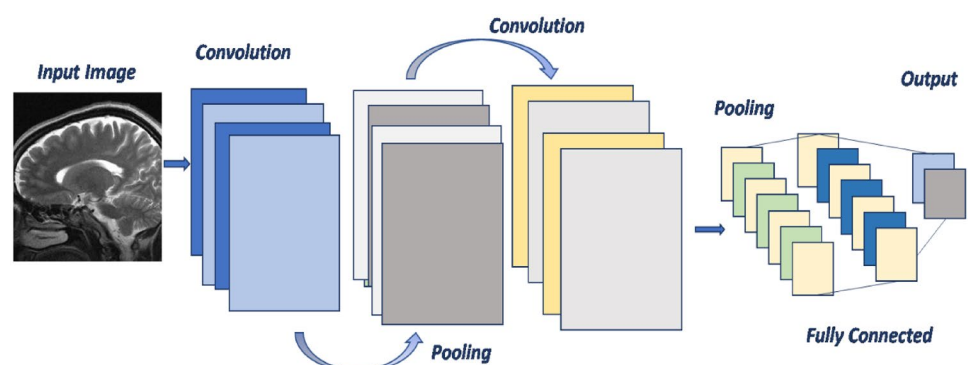
Features: These are labeled features in the input image. **Weights:** Adjustable parameters that determine the influence and strength of each feature on the neurons in the hidden layers. **Hidden Layer:** These are layers of neurons that process input features based on weights. This is how models learn complex patterns within the data.

In our context of liver cancer diagnosis, we have presented briefly the strengths and limitations of ANNs:

- **Strengths:** ANNs excel at learning from large datasets, making them suitable for applications requiring pattern recognition, such as liver cancer detection. ANNs can model complex, non-linear relationships, making them applicable for many applications in liver cancer diagnosis, such as predicting cancer risk based on patient data.
- **Limitations:** The performance of ANNs depends heavily on the availability and quality of large datasets. In liver cancer diagnosis, acquiring extensive labeled datasets can be challenging. ANNs are prone to overfitting, especially with limited data, which can lead to reduced generalization and accuracy in diagnosing new, rare (unseen) cancer cases.

3.2.2 Convolutional neural networks (CNNs)

The use of CNNs has gained popularity significantly because of their impressive feature extraction. CNN is a deep learning algorithm based on the animal visual cortex and therefore is used widely for imaging-related data, such as CT and MRI. They have been designed to automatically learn features from images. CNNs are used because, with their structure, they are able to accurately extract relevant features from input data without the need for traditional methods like texture analysis. They also don't require segmentation of images for training.

Fig. 4 Convolutional Neural Network

CNNs are made up of 3 layers: convolution layers, pooling layers, and fully connected layers. The first 2 layers are involved in feature extraction, and the fully connected layer maps the extracted features out as the final output, thus classifying the feature [43, 45]. As illustrated in the figure below, CNN architecture is used to perform object detection in an image task. Figure 4 shows the layers and architecture of a CNN model and below we have explained the functionalities of each of these layers.

Input: The image being analyzed. Convolutional + pooling layers: These layers act as feature extractors. Each convolution + pooling layer narrows down and the features selected get more precise. The more features and detailed the output can be, the more such layers. Fully connected layers (FC): The FC layers act as classifiers. Probability values are assigned to the features extracted, and then FC layers predict the class in which the image belongs.

In our context of liver cancer diagnosis, we have presented briefly the strengths and limitations of CNNs:

- Strengths: MRI and CT imaging are the most efficient, cheap and safe ways of diagnosing and screening liver cancers. CNNs excel in extracting relevant features from medical imaging data without requiring manual segmentation, making them highly accurate and crucial in this context.
- Limitations: Training and using CNNs does, however, require computational resources not readily available in all clinical settings. This makes them difficult to implement. The decision-making process of CNNs can be opaque, making it difficult for clinicians to interpret the results and understand the rationale behind certain diagnoses. This is vital information, especially when diagnosing life-threatening diseases like cancer, as it can help in prognosis and patient well-being.

3.2.3 Fully convolutional networks (FCNs)

FCNs are a modification of a CNN, in which fully connected layers are converted into convolutional layers, and deconvolutional layers are added in order to attain an output map from the original image [46]. FCNs are majorly used for segmentation, or more specifically, semantic segmentation. Semantic segmentation is a deep learning algorithm in which every pixel in an input image is assigned a label [47]. This way, the algorithm forms collections of pixels within the whole image, isolating clearly all separate features. FCNs are extremely beneficial for feature extraction for this reason.

In our context of liver cancer diagnosis, we have presented briefly the strengths and limitations of FCNs:

- Strengths: Semantic segmentation mentioned above is highly useful in cancer diagnosing through imaging. It enables FCNs to conduct a detailed identification of cancerous regions within liver images at the pixel level. This way, FCNs also automate the segmentation process, which saves time and reduces the potential for human error in manual segmentation.
- Limitations: Similar to CNNs, FCNs require large annotated datasets for training to achieve high performance, which can be a limiting factor in liver cancer diagnosis. Computational complexity is also high for FCNs, questioning their clinical applicability.

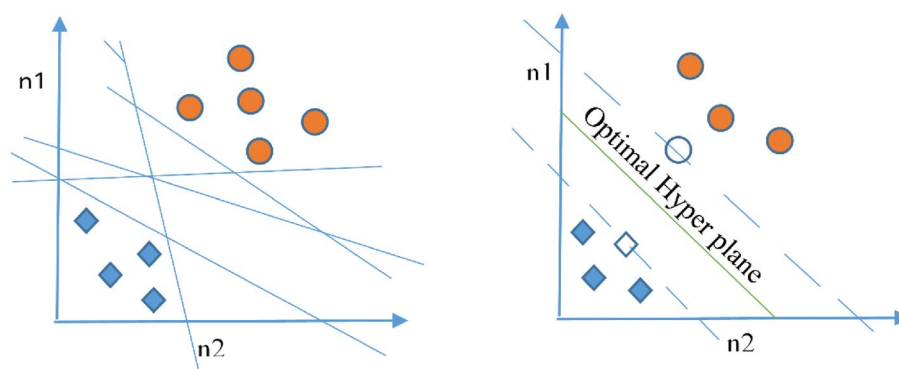
3.2.4 Support vector machine (SVM)

SVMs are a classification tool and a supervised learning ML algorithm used in classification tasks. Similar to other algorithms, SVMs are trained by exemplar data, and can then use this knowledge to make new decisions. Specifically, SVMs are trained to assign labels to objects. Because SVMs inherently work with linear functions, they work best and most commonly for binary decision-making and classification; however, modifications can be made to change this. SVMs are put at the end of AI models, where segmented images and highlighted features are input into the SVM for it to classify the object—in our case, the type of liver cancer [48]. Figure 5 shows the output of a SVM model for a binary classification problem. This is further explained in the below paragraphs.

The blue and orange data points signify opposite classes. Maximum Margin Hyperplane: The optimal hyperplane that is equidistant between the positive and negative hyperplane. The goal of this classifier is to maximize margins, therefore making clear distinctions between the two classes. Positive Hyperplane: The boundary before the positive class data points. Negative Hyperplane: The boundary before the negative class data points. Support Vectors: These are data points closest to the optimal hyperplane (shown in white) and decide the position and orientation of the hyperplane.

In our context of liver cancer diagnosis, we have presented briefly the strengths and limitations of SVMs:

Fig. 5 Structure of an SVM algorithm



- Strengths: SVMs are effective even when working with smaller datasets, as they can still perform well in binary classification tasks related to cancer diagnosis. This makes them a good choice for most cancer diagnostic models. SVMs can also effectively handle high-dimensional data, making them useful for complex feature sets taken from imaging and patient data.
- Limitations: Although SVMs are extremely effective in binary classification, liver tumors have variants including different cancer types and benign tumors. This may limit their applicability in non-binary multi-class cancer diagnosis without significant modifications. Also, SVMs often require manual feature selection, making them more labor-intensive and creating the risk of not capturing all relevant features when diagnosing.

3.3 Transfer learning models

Transformers are a form of deep learning models that have started to rise in use of medical diagnosis. Transformers are popular as they utilize self-attention mechanisms to enable themselves to weigh the relative importance of different parts of the training and input data. This means that they are able to interpret what data and features are more significant for making a final decision, such as a diagnosis. Self-attention mechanisms enable transformers to capture long-range dependencies—relationships between elements of the data that are spread apart—unlike traditional neural networks. Similarly, transformers excel in utilizing global contextual information within datasets. To exemplify this in our context, in a medical image, accurately segmenting an organ requires understanding the overall structure and relation of the organ to surrounding tissues and other organs. This is a benefit of using transformers [49, 50]. Below, we have elaborated on commonly used transformers in the diagnosis of cancers using AI.

3.3.1 Medical transformer (MedT)

MedT is a specialized Transformer type designed for medical image segmentation. As mentioned above, as it is a transformer, MedT leverages self-attention mechanisms, allowing it to capture long-range dependencies and global contextual information within images and other data. In our context of liver cancer diagnosis, we have presented briefly the strengths and limitations of MedT below [51]:

- Strengths: It can capture long-range dependencies and thus can effectively model global contextual information to capture anatomical structures and subtle features essential for accurate cancer diagnosis and differentiation (crucial for liver cancers). Transformer architectures, including MedT, are highly scalable. They can handle large datasets efficiently, making them suitable for large medical screening tasks.
- Limitations: MedT requires significant computational resources due to the complexity of Transformer models. This can limit its application in real-time clinical settings. Additionally, the training of Transformer-based models typically requires large, annotated datasets. In medical data, obtaining such datasets can be challenging due to the lack of annotated medical images.

3.3.2 U-Net variants

U-Net is a type of CNN architecture designed especially for the segmentation of medical images. Over time, as they have gained popularity, various U-Net variants have been developed to enhance performance for specific tasks. They are mainly known for the simplicity and efficiency they add to the process of medical imaging. U-net is also special in terms of handling similarities between medical images. The model employs an encoder-decoder structure with skip connections that facilitate the transfer of detailed information from the encoder to the decoder, thus improving segmentation accuracy and its ability to identify unique images [52]. In our context of liver cancer diagnosis, we have presented briefly the strengths and limitations of U-Net below.

- Strengths: U-Net and its variants are generally simpler and less resource-intensive than Transformer-based models. This makes them easier to train and deploy even with fewer computation resources. Also, it performs well even with limited training data, which is common in medical applications. U-Nets have a proven track record of effectiveness in a wide range of medical imaging tasks, including liver cancer detection. Their consistent performance makes them a reliable choice for many applications.
- Limitations: U-Nets primarily focus on local features, which can be a limitation when capturing global context and long-range dependencies. This can affect their ability to accurately segment complex structures compared to Transformer-based models. They also sometimes struggle with larger datasets and high-resolution images.

3.3.3 TransUNet

TransUNet combines the strengths of U-Net, a known CNN-based architecture, with the capabilities of Transformers. This hybrid approach integrates a Transformer into the U-Net structure, improving the model's ability to represent features by efficiently using both local and global contexts. This ability significantly improves model precision and differentiation capabilities as it aids in the identification of tumor regions in images [53]. For example, a study using TransUNet for liver tumor segmentation reported a significant improvement in segmentation accuracy compared to traditional U-Net models [54]. It also aids in delineation of tumor, reducing manual load for users.

- Strengths: The hybrid structure of TransUNet gives the model the strong local feature extraction of CNNs with the global context understanding of Transformers. This helps have good precision while also a spatial understanding of the cancerous region, in our case the liver parenchyma. TransUNet can be tailored to different medical imaging modalities, making it more applicable in clinical scenarios irrespective of availability of imaging tools.
- Weaknesses: The integration of Transformers into U-Net increases the model's complexity, which can complicate training and deployment. This can make implementation difficult and expensive.

3.4 Fuzzy logic for AI-based diagnostic systems

Fuzzy logic is another computational technique used when creating AI-based diagnostic models. The purpose behind Fuzzy logic is to better replicate the uncertainty behind cases in the real world. Unlike Binary or Boolean computing, in which 0s and 1s are used to encode all outcomes, Fuzzy logic allows us to create different “degrees of truth”.

When dealing with complex data, Fuzzy logic is commonly used to replicate human thinking and cognition. It can be extremely helpful in handling machine outcomes based on numerous input parameters. In the context of deep learning, Fuzzy logic can enhance model decision making by allowing them to handle imprecise inputs and make decisions based on degrees of truth rather than binary decisions. More specifically, liver cancer detection using deep learning often involved analyzing complex imaging data, where variations in tumor appearance and boundaries between organs can make it difficult to conduct binary classification on the type of cancer. Integrating Fuzzy logic with deep learning can allow for more nuanced decision-making, providing probabilities rather than absolutes—leading to improved sensitivity and specificity for liver cancer detection. For instance, Fuzzy logic can help identify the precise stage of a cancer, or the likelihood for a particular case [55–57].

4 Literature survey

The below subsection first shows a comparative analysis between popular imaging techniques used for training AI models. Furthermore, we provide reasoning behind the potential of 3D imaging datasets. This review also consists of original studies analyzed, which were chosen based on the inclusion criteria shown above. 8 of the above papers contain some variants of CT datasets, as it is the most popular choice in AI models. 2 papers contain MRI datasets, and then miRNA followed by DNA methylation dataset based studies have been analyzed. The difference in dataset types depicts the ratio observed during research.

4.1 Imaging for deep learning AI models

In the past, the most efficient non-invasive method for liver tumor diagnosis has been some form of imaging (one of the 3 previously explained) [58]. Imaging is used because it can give information about the precise location of the tumor/mass and its size and shape. This is also extremely important in being able to differentiate between types of liver cancers. It's shown that there are particular important features in imaging that are more likely to appear in HCC than in ICC, for example. These include intralesional fat, lesion washout and the presence of a capsule [59].

This shows that accurate and efficient imaging is extremely useful for diagnosis of liver tumors. Differentiating between types (HCC vs ICC) also gives information for the prognosis of the tumor.

When training Deep Learning AI to diagnose and classify liver masses, CT and MRI are the preferred choices for input training datasets. Ultrasonography (US) is fast, but has other disadvantages that make its use limited: US is unsatisfactory because of its low sensitivity to small liver masses and the dependency on a manual US operator [60]. CT and MRI are both automated and done through machines. The two also provide a higher resolution and level of detail—enabling higher precision in distinguishing normal and abnormal tissue/patients [61]. According to research by Zhen et al. [60], enhanced MRI is the best tool for diagnosing liver masses because it has various mechanisms for tissue contrast and has the ability to clearly check auxiliary imaging features. MRI also allows good categorization of small tumors (≤ 3 cm) [61]. Even with this, MRI is not used widely unlike CT and US because of the high cost and time associated with it.

Additionally, MRI and CT can be combined with contrast agents to assess blood flow in the liver, which is an important factor to help distinguish benign and cancerous lesions. Digging deeper, it was observed that between these two (MRI and CT), CT was the most opted-for option. Alongside the high resolution, another advantage of the use of CT is that it allows for similar or better evaluation of extrahepatic tissue, including bones, bowel, lymph nodes, etc. as MRI while being quicker and more affordable [62]. It is therefore recognized as the preferred medium for imaging cancers in the liver despite the ionizing radiation exposure to the patient [61]. Because of this, most of the studies analyzed use a CT-orientated dataset.

4.2 2D and 3D datasets for deep learning AI

Both CT and MRI scans are inherently 3D, but CT is often represented as a continuous series of 2D cross-sectional slices of the 3D scan. This is done to make it easy for radiologists to interpret the data. Also, 2D data requires significantly less computational complexity, power, and storage, thus is the preferred choice for the localized analysis of a disease region where a single cross-sectional view is sufficient. However, 2D images create a lot of restrictions. They reduce our ability to understand spatial analysis, where experts check for the spatial relationships between structures in the body. 3D images give a more holistic view and help to view anatomical structures more clearly. In 2D images, overlapping structures are harder to differentiate and identify as well.

With time, a new trend has come where 3D data is being used more often for training and testing AI models for diseases. For CT scans, a 3D dataset is created by stacking 2D slices, resulting in a volumetric representation of the imaged area. In the case of cancers, a volumetric representation and a 3D view make the shape and location of the tumor more easy to locate. 3D data also helps monitor tumor growth and vascularization. For example, they often incorporate blood vessel distribution around the tumor, which helps in differentiating cancers.

Moreover, 3D datasets are definitely more comprehensive in comparison to 2D; however, algorithms incorporating 3D data tend to be significantly more complex, increasing training and testing time and computational complexity. This may reduce a model's ability to be used in day-to-day clinical applications.

4.3 Fuzzy logic-based diagnosis for liver cancers

Fuzzy logic has a lot of potential in the field of liver disease diagnosis. Ahmad et al. [63] showed that fuzzy systems can be used to accurately classify liver diseases. Ahmad et al. [63] proposed a multilayered Mamdani fuzzy inference system for diagnosing a liver virus (Hep-B) using a fuzzy-based inferential system. Similar to the potential fuzzy logic applications listed earlier, this model is able to classify different phases hepatitis B because of fuzzy logic. The system showed a 92.2% reliability. Furthermore, [64] uses a Mamdani fuzzy inference method to create a 95% accuracy model for liver fibrosis detection. This study explained that fuzzy logic can be used to deal with inaccuracies and vagueness in patient historical data. In addition, by using fuzzy data it was possible to classify patients' health into different classes. Mirmozafari [65] utilizes an expert system for diagnosing liver disease. These systems are similar in function to Fuzzy systems as they store experts knowledge and use it to process information and make decisions (generate outputs). Lastly, another study, [66], made use of a Fuzzy Cluster Means (FCM) system to diagnose liver diseases. Using FCM aided in evaluation and classification of patients into multiple classes of liver diseases.

The integration of fuzzy logic into liver disease diagnosis demonstrates significant potential in increasing accuracy and reliability. Various studies above demonstrate its effectiveness. Fuzzy logic allows for impressive classification as it becomes easier to identify stage of disease and differentiate between classes of diseases/cancers. Fuzzy-based models certainly have potential in liver cancer diagnosis; however more research in the field is needed.

4.4 Research analysis

This subsection highlights key research studies on liver cancer detection and classification using a variety of dataset types and deep-learning AI models. Table 2 provides a description of different studies along with their sample size, case and objective.

Table 3 presents the research analysis done on recent studies. Here, we have analyzed the challenges associated with the various Deep Learning algorithms used in the proposed models, while also noting their future scope in medical practices.

5 Benchmark AI-based framework

The ability of artificial intelligence ML and DL to make complex decisions based on empirical data presented to them has shown to be significantly helpful in modeling and aiding in complex problems like liver cancer and other diseases. AI tools are constructed following specific standard processes to work efficiently for any application. These tools then aid in faster, more efficient and reliable diagnosis and prediction of diseases. A benchmark construction procedure of AI-based diagnostic systems is shown in Fig. 6. This benchmark framework aims to address commonly occurring limitations in previous works, such as balancing, overfitting, etc. This structure is not a complete model, rather, can serve as a reference point for future models and researchers can build off of this. This is important as it addresses common limitations.

Exploratory data analysis (EDA) refers to the data exploration process. It is used to better understand the information presented in datasets by highlighting important characteristics and is used to spot patterns and relationships between variables. Techniques such as correlation analysis, filtering out non-essential data, and identifying outliers are commonly used to extract meaningful insights from the data. Next, we go on to preprocessing techniques.

First, *Data Curation*. In this step, data is managed and organized by the curators to make it easier for users to analyze the data. Further investigation of the data is done through analysis of associated repositories. *Class balancing* is then done to the dataset when the instances of each class are not equally distributed within the dataset (i.e. class imbalanced data). The imbalance Ratio (IR) is computed using the size of the minority class (M_i) and the size of the majority class (M_a). IR is calculated using the Eq. (1).

$$IR = \frac{M_i}{M_a} \quad (1)$$

This is extremely important to acknowledge because training AI on such data will make the model biased towards the majority class, and show poor performance for the minority classes. Oversampling (increasing the number of instances in the minority class synthetically or by duplicating) and undersampling (decreasing the number of instances in the majority

Table 2 Research analysis

Authors	Dataset type	Sample size	Cases	Objective
Zhen et al. [60]	Contrast-enhanced MRI, unenhanced MRI	31,608 images	1210	Dynamic contrast-enhanced MRI allows for the most holistic and comprehensive observation for differential diagnosis of liver tumors. However, the subjectivity of doctors poses a problem. Deep learning is used to combat this. Assisted diagnosis of liver tumors by categorizing them into 7 categories using enhanced, and unenhanced images and clinical data (like text)
Gao et al. [71]	Multi-Phase Contrast-enhanced CT	Not mentioned	499	To create an AI diagnostic assistance system for doctors to help differentiate malignant hepatic tumors based on multi-phase CECT images
Wang et al. [72]	Dynamic CT	Not mentioned	234 (157/77 for train/test)	To integrate the siamese mechanism into (cross-contrast NNs) CCNNs, allowing them to predict like traditional CNNs, while taking advantage of the siamese twin network mechanism for better pattern recognition for diagnosis of HCC and ICC
Yang et al. [73]	Contrast-enhanced CT	Not mentioned	Training/testing—198/85	To aid in prognostication of HCC recurrence, recurrence-free survival (RFS) and overall survival (OS) of early-stage HCC patients through the use of radiomics
Yasaka et al. [67]	Dynamic CECT	1068 images pre-augmentation. 55,536 images post augmentation	460 patients (for training). 100 patients (testing)	To develop a CNN to analyze dynamic contrast-enhanced CT scans and automatically classify liver masses into distinct categories (HCC, Malignant tumors other than HCC, indeterminate masses, hemangiomas, cysts). The study aimed to overcome previous limitations in the field in regard to differentiating indeterminate lesions. One example of this is the similarity in attenuation of early HCC and dysplastic nodules leading to false diagnoses)

Table 2 (continued)

Authors	Dataset type	Sample size	Cases	Objective
Das et al. [74]	CT	225	75 patients (46 male, 29 female)	The study aimed to develop a new automated method for liver cancer detection in CT images. Thus, an automated CAD system was developed for decision-making processes for cancer diagnosis. New technique introduced: Watershed Gaussian-based deep learning (WGDL) technique. For cleaner images and improved segmentation and classification
Rajkumar et al. [75]	CT	5	Not mentioned	The primary aim of the study was to develop a more accurate method for automatic liver tumor detection in CT images by combining two popular methods, KBC methods for segmentation and SVM classifier
Chlebus et al. [76]	CECT	131 CECTs	Not mentioned	This study aimed to create a U-net based FCN model with object-based postprocessing for successful and automated liver tumor segmentation in contrast to enhanced CT scans. The model aims to utilize full 3D volumetric information for more efficient classification of tumors
Ben-Cohen et al. [77]	CT	NA	20 patients with 68 lesions (training). 14 patients with 55 lesions (testing)	The research aimed to develop a novel model for the detection of liver lesions, specifically for increasing successful diagnosis and detection of small lesions and metastases in CT scans
Trivizakis et al. [68]	Diffusion-weighted MRI	130 DW-MRI scans	Not mentioned	This study proposes a novel and rather unpopular idea of utilizing 3D CNNs as feature extractors and SVMs as classifiers for diagnosis of liver cancers in a diffusion-weighted MRI dataset. The study aimed to combat difficulties in previous studies in distinguishing between primary and metastasized tumors in the liver when using 2d CNNs

Table 2 (continued)

Authors	Dataset type	Sample size	Cases	Objective
Sayed et al. [69]	Micro RNA (miRNA)	Not mentioned	Not mentioned	The study's primary objective was to identify 3 novel miRNA biomarkers that could be used as early diagnostic markers for HCC. The study utilizes these biomarkers and their expression profiles to propose a highly accurate machine learning model that predicts HCC based on miRNA profiles of patients. The model is robust as it also incorporates hematological, biochemical, and microbiological markers
Bai et al. [70]	DNA Methylation sites	96 + 402 + 108 liver samples	Not mentioned	The study aims to identify DNA methylation markers for 2 purposes: differentiating primary liver cancer (PLC) with normal tissue, and differentiating between HCC and ICC. The study develops machine learning models that differentiate between the 4 categories above with high accuracies based on methylation data of the patient
Liu et al. [78]	mRNA, lncRNA, miRNA, CNV, and SNP	320 + 320 + 321 + 324 + 313. 302 samples for training	Not mentioned	The study aims to develop a more accurate prognostic model for HCC using multi-omics data. This is done by integrating various omics data, including mRNA, lncRNA, miRNA, CNV, and SNP information. These datasets were used since they clearly represent biological phenomena behind HCC
Lu et al. [79]	Gene expression profiles	225 + 268 + 166 + 48 + 91 = 798	Not mentioned	The goal of this study was to identify key pathways and genes involved in the pathogenesis of HCC through cross-species comparison and bioinformatics analyses

Table 2 (continued)

Authors	Dataset type	Sample size	Cases	Objective
Schmauch et al. [80]	Ultrasound	367 images	Not mentioned	The study developed a deep learning model for the detection and classification of focal liver lesions into the following categories: malignancy, angioma, hepatocellular carcinoma, focal nodular hyperplasia, cyst, or metastasis—done using ultrasound imaging data. It was designed to perform two main tasks: detecting abnormalities and classifying lesions as benign or malignant
Suyama et al. [81]	FDG-PET vs. EOB-MRI scans	Not mentioned	34 liver metastasis cases	The study investigates the diagnostic ability of a deep learning image reconstruction (DLR) method in detecting liver metastases cases using FDG-PET imaging. The primary purpose of the study was to compare the efficacy of the DLR in liver metastases diagnosis against the established EOB-MRI findings
Kanan et al. [82]	CT	Not mentioned	121 patients with liver metastases	The primary objective of the study was to determine whether Deep Learning Image Reconstruction (DLIR) can aid in the detection of liver metastasis cases in the CT scans of patients. The detection was performed by radiologists who utilized both DLIR and a more traditional adaptive statistical iterative reconstruction (ASIR-V) method. The conspicuity of the 2 were compared to determine which method was better
Shen et al. [83]	Gene expression profiles	NA	362 patients with HCC	The primary goal was to identify a unique gene signature that could predict the recurrence of HCC in past-HCC patients. The study aimed to create a cost effective gene panel for clinical use since whole-exome and whole-genome sequencing is expensive. The ML model was made after analysing gene mutations from databases and comparing that with whole-exome sequencing of 10 patients

Table 3 Description of challenges encountered and the possible future outcomes for studies analyzed

Authors	ML/DL model used	Challenges	Future scope
Zhen et al. [60]	CNNs	High number of false negatives for HCC diagnosis. Radiologists performed visibly better for the category metastatic malignant tumor	Adding a variety of unique cases in the training set will help reduce false negatives. Better utilization of clinical information will help for improved metastatic malignancy categorization. “accurate and time-saving assisted-diagnostic strategy for liver tumors”. The use of unenhanced images in training allows for use in the absence of contrast agents
Gao et al. [71]	STIC model with integrated deep CNN and gated RNN	Small training set and thus difficult for atypical cases and generalizability. Increased training time because of multi-phase CTs. The model alone had a low accuracy (ranging between 70–80%) but STIC-assisted doctors set showed high accuracy	Since the STIC-assisted doctors set showed a significant increase in accuracy and sensitivity in the diagnosis of malignant hepatic tumors (26.9% increase in sensitivity for ICC), the model has scope for application in clinical settings as a CAD tool that is relatively easy to interpret
Wang et al. [72]	Siamese cross contrast neural network (SCCNN)	The performance of SCCNN is influenced by the choice of the base model, and more complex base models may not always lead to better results. SCCNN feature extraction is distributed throughout the entire liver, showing increased reliability in non-lesion areas. This hinders diagnosis of larger tumors	SCCNN is a viable choice in situations where less medical data is available. The model shows good performance as it is able to extract more discriminative features and work with lesser medical data. This shows its ability to be a CAD tool in clinical applications
Yang et al. [73]	Radiomics. Pretrained CNNs, SVM classifier	Retrospective study, therefore, training and testing was done on past data, thus making generalizability in present times difficult. Small dataset as well	Model showed satisfactory MVI (microvascular invasion) prediction and high efficiency overall. The model also doesn't require manual delineation. This automation and accuracy allow it to aid in clinical decisions on patients with early HCC and to help in the recurrence of tumors and even mortality
Yasaka et al. [67]	CNNs	The model showed low sensitivity for diagnosis of indeterminate lesions including early HCC and dysplastic nodules likely due to the subtle difference shown in the slices and limited training data for those masses. The sensitivity for rarer masses was lower for the same reasons. Models like CNNs also experience low performance because of overfitting caused by large datasets	The model showed high performance regardless of what was shown around the liver masses. This might have been achieved by training with several liver masses in various sections of the liver and because of high augmentation of image sets. Additionally, the model showed high performance for most masses encountered in daily clinical practice, making it applicable in real-life settings
Das et al. [74]	Watershed Gaussian based deep learning. DNN classifier	Use of watershed transform and Gaussian mixture model makes the model have high computational complexity, making it difficult for regular clinical applications. Small dataset. The model also struggles with calculating the volumetric size of lesions	The model showed a 99.38% accuracy and 100% sensitivity in detecting liver cancers in the testing set. This great success makes the model viable for use in clinical diagnosis and decision-making process for diagnosis of liver cancers. Furthermore, making 3-D mesh structures from different slice of the images will help calculating volumetric size of the image

Table 3 (continued)

Authors	ML/DL model used	Challenges	Future scope
Rajkumar et al. [75]	SVM, KBC (knowledge-based constraints)	Extremely small test size of only 5 images. SVM classifier is not optimal because of the large datasets. The proposed model also requires excess manual interaction and has a slow computational speed	The model has extremely high accuracy of above 98% and above 98% sensitivity and specificity. The model should be tested with a larger set to check generalizability and be trained with more images as well
Chlebus et al. [76]	2D FCNN and Random forest classifier (RFC)	The use of voxels to utilize volumetric information led to the high false positive rate in the study. The model also showed a low success rate for detecting smaller tumors, and an overall inferior detection (not segmentation) success in comparison to humans (63% vs. 92%)	The model gave results comparable to a human expert (77% detection), however, is capable of detecting bigger lesions with better accuracy than smaller ones. More 3D volumetric information should be utilized for better accuracy
Ben-Cohen et al. [77]	FCN, Sparsity-based classification	The FCNs performance was worsened because of slice thickness of the CTs. Along with this, the use of the automatic liver segmentation tool reduced the true positive rate slightly. Inaccuracy of this tool caused some small proximal detected lesions to be merged	Higher resolution and thinner slices would improve the detection rate, especially for small lesions. The model showed a fully automated convolutional network with sparsity-based false positive reduction for the detection of liver lesions, specifically liver metastases in CT. The model presented a high 88.2% true positive rate, and a reduced false positive rate with sparsity classification. This makes it viable for trial in clinical applications
Trivizakis et al. [68]	3D CNNs, SVM classifier	In DW-MRI, primary and metastasized tumors exhibit slower signal decay at high b-values, giving this study a low accuracy when distinguishing between the 2. Small dataset. 3D CNNs increased computational complexity making clinical applications difficult	When compared to 2D CNNs, which were similarly trained, the model displayed a higher accuracy for all tests (85.5%/71.9% for primary vs. metastasized tumors for 3DCNN/2DCNN). The model also removes the need for processes like delineation, cropping, and ROI annotations reducing time and making it viable for clinical application if computational complexity is reduced
Sayed et al. [69]	BAVO (binary African vulture optimization algorithm) and SVM	Firstly, the model was trained and tested on a small dataset, as acknowledged by the authors, and this may affect the generalizability. Also, the dataset only uses Egyptian patients' profiles, reducing diversity in the training data. Next, out of the 3 miRNAs tested, 2nd and 3rd miRNA showed low accuracy and performance due to the limited available information on them. This shows a limitation of miRNA datasets since they are still undergoing research and information is not readily available for many miRNAs	The model should be tested and trained on a larger cohort to make it applicable in clinical use. However, the accuracies for all 3 miRNA studies were high in comparison to similar studies. This study therefore encourages research on the miRNAs found to be linked to early-HCC. Lastly, the novel BAVO algorithm introduced in this paper for an efficient feature selection and optimal parameter selection has lots of potential implications in the field as it resulted in high accuracies in this paper

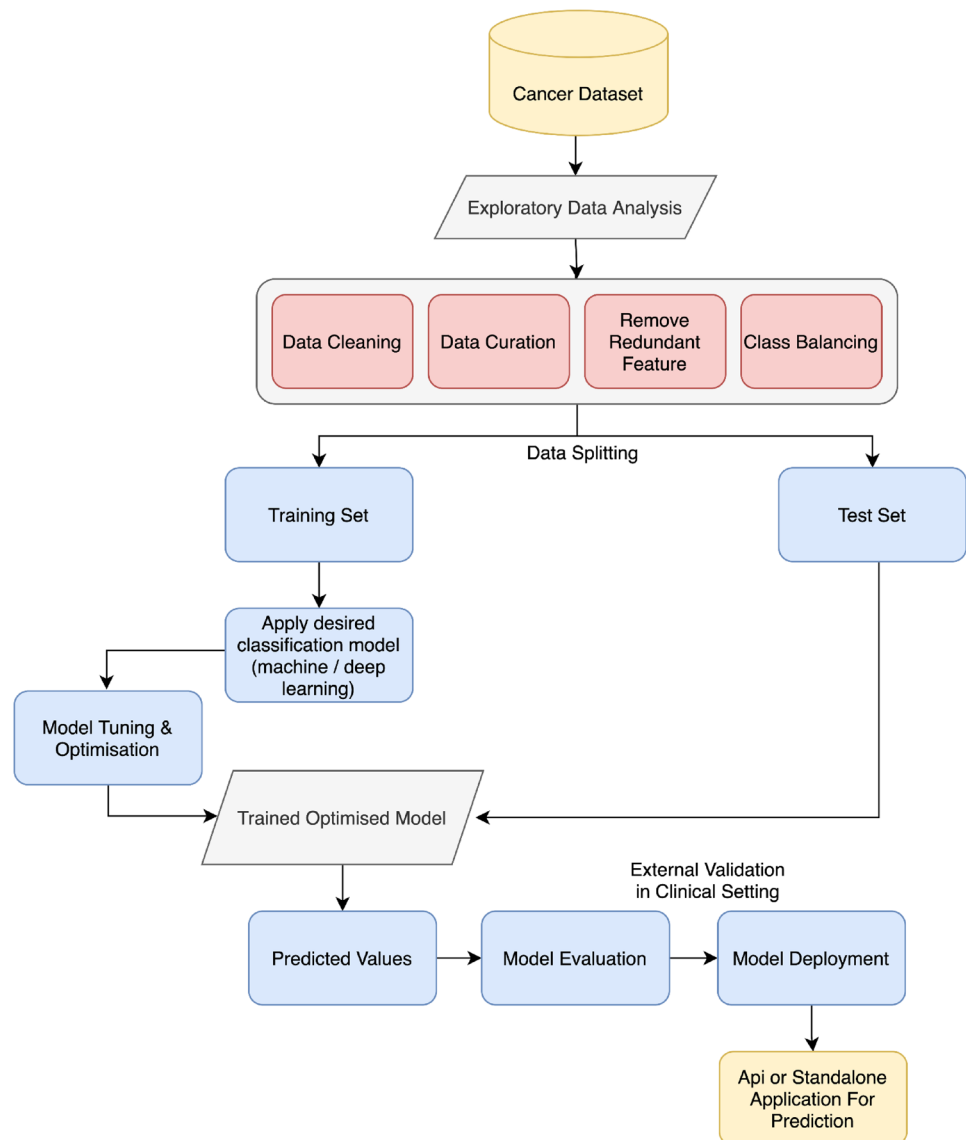
Table 3 (continued)

Authors	ML/DL model used	Challenges	Future scope
Bai et al. [70]	Random forest	The most significant challenge in the study was its dependency on biopsies, which are traumatic to the patients. This means that for clinical applications of this model, taking relevant tissue samples from patients will be traumatic and thus not the best approach. The invasive approach utilized is the challenge, and so the efficacy of the methylation biomarkers may not be the same in actual patients	The problem of differentiating between cancer types in the liver is a big one (because its prone to metastases), but the use of methylation biomarkers showed great accuracy for classifying HCC, ICC and PLC in the liver. This could also be used to differentiate metastatic tumors, cysts and lesions from PLCs as well, resolving a common problem with diagnosing liver cancers. Since the authors plan to test the model again on peripheral blood samples, the trauma associated with this method of diagnosis may be resolved with future research
Liu et al. [78]	Multi-omics	There was no external validation for 3 of the single-omics models due to the lack of independent datasets. Similar problem persists for the multi-omics model as well. Next, the final multi-omics model includes 30 + biomarkers and thus may be difficult to use in clinical setting. The success of the model strongly depends on patient willingness to undergo sequencing	The multi-omics model exhibited better performance (Concordance index based) than all single-omics models. Multi-omics data has great potential in terms of accuracy and prognostication of liver cancers. External validation is necessary for the model to go for clinical use. As mentioned earlier, simplification of the model is necessary to enhance practicality for patient comfort and easier usage
Lu et al. [79]	Gene set enrichment analysis (GSEA) and meta-analysis	The study acknowledged that there was a small sample size, which could lead to unreliable variance estimation and thus a higher false positive rate. This may also cause accidental overlooking for some significant differences in gene expression levels of different samples. The use of different samples and experimental platforms may cause the datasets to be unstandardized, potentially reducing the robustness and generalizability of the data	The study identified the cell cycle pathway as significantly enriched across multiple datasets. Specifically, cdc25a was noted as a key gene in HCC tissues compared to normal tissues across the 3 species. Further validation is required for the identified pathways; however, these have diagnostic potential for HCC. Integration with a machine learning model may allow for genome-wide data interpretation leading to more identified pathways linked to HCC
Schmauch et al. [80]	ResNet50, supervised attention mechanism	There was a clear imbalance in the distribution of the lesion types within the dataset, with some (ex: HCC) being underrepresented. This limited exposure to diverse cases reduces the generalizability of the model. Additionally, the author's acknowledgment that some annotations may be unrepresentative of the true nature of lesion in the ultrasound image: labels such as "HCC" and "Metastasis" don't make sense to provide as they are difficult to diagnose with ultrasound alone. A larger testing dataset is also required to validate the model	The model uses a novel approach to limit the annotation time for radiologists. Before further implementation, traditional more precise annotation methods should be tested to see if they result in stronger accuracies. Simultaneously, the new annotation approach here also may have implications for other imaging-based deep learning models. Moreover, a larger external validation set should also be considered, which was also acknowledged by the authors of the study

Table 3 (continued)

Authors	ML/DL model used	Challenges	Future scope
Suyama et al. [81]	Deep learning image reconstruction (DLIR)	The study only involved 34 diagnosed cases, not making the results very reliable. Additionally, although the DLIR method certainly showed a higher diagnostic score, the p value was only 0.14, indicating that it was not statistically significant. Lastly, the EOB-MRI findings showed some nodules that were missed by the DLIR, indicating a lower sensitivity	Through the low p value and possibly low sensitivity, it is unjustified to say that DLIR is significantly a better approach to diagnose liver cancer metastases; however, the research does show potential for DLIR methods in the future, as they did increase the diagnostic score in the study. Larger datasets and improvements on the sensitivity should be points for further research
Kanan et al. [82]	Deep Learning Image Reconstruction (DLIR)	This was a retrospective study that used only a single CT and only 1 manufacturer's DLIR. This makes it difficult to generalize the results. Additionally, the inclusion criteria for the data used added limitations to the study: the study only included patients with liver metastases, meaning that the results only show DLIR to improve sensitivity, not specificity	The high-strength DLIR statistically increased the detection and conspicuity of liver metastases compared to ASIR-V. This shows its implications in improving differentiation of PLCs and SLCs. Being used as a pre-processing tool, DLIR is easier to implement in clinical scenarios as it is not fully automatic; rather, it is an assistive tool for radiologists
Shen et al. [83]	Decision tree versus SVM	The model's verification on a small sample size of 10 patients limits the generalizability. Larger validation set is needed, given the complexity and heterogeneity of HCC. The study noted differences in gene mutations between races (Asian vs non-Asian population) which could affect the model's applicability in real-world scenarios with diverse patients	The 2 models tested showed accuracies of 74.19% (decision tree) and 80.65% (SVM). Although the SVM showed higher accuracy, the authors noted that the decision tree model is more interpretable and thus should be focused on to improve in future research as understandable models have greater potential for clinical implementation. Additionally, the gene signatures need to be validated with a larger, multi-ethnic cohort to confirm the performance. Multi-omics approach can also be taken to extend this research by combining with other data, such as miRNA

Fig. 6 Benchmark AI-model framework



class to match the minority class) are two major techniques used to achieve class balancing. Hybrid techniques can also be used. Another preprocessing technique done to the dataset is *removing redundant and noisy features*. Because these add unnecessary load to the training data when training an AI-based model. This reduces the model's accuracy because relevant features are overshadowed, and also reduces its generalizability.

Data standardization is a process used to standardize the independent features (columns) in the dataset into the [0, 1] interval. It reduces the skewness of the data distribution by shifting the individual attributes to have an average of zero and unit variance. The standardization (Z_n) is attained through the z-score formula below, Eq. (2).

$$Z_n = \frac{(n - \mu)}{\sigma} \quad (2)$$

Here n represents the data instance while μ and σ are the mean and standard deviation of the feature respectively.

Lastly, *data splitting*: This is a step done to measure the performance of the model developed. The dataset is split into 2 parts, where one is used to train the model and the other is used to test the model. This step evaluates the performance of the model on unseen, new data.

Then we can go ahead to add the classifier. This step is highly variable depending on the type of data used. Different models are chosen for their performance in different scenarios. For example, neural networks require large datasets and training. The choice of the classifier is critical as it will determine the performance. Then we *tune* the classifier—selecting proper hyperparameters in order to maximize the classifier's performance. *Optimizing* hyperparameters is also done to improve the performance of the model overall. For example, Grid search to find optimal hyperparameters combination for the best diagnosis and classification. The performance of the model is then measured and *evaluated* on various parameters, such as accuracy, sensitivity, etc. A more elaborate explanation of these parameters can be seen in Sect. 6.1. Although some parameters like accuracy are usually applicable, the choice of parameters chosen also depends on the class balance within the dataset.

Model deployment refers to the process of integrating a trained DL/ML prediction model into a productive environment as a decision-making diagnostic assistant. The deployed model is then validated, in terms of performance, through external means (such as clinics and other medical facilities). Once sufficient validation is done and the model is proven to be useful and accurate, it can then be utilized in general clinical practice. Once validated, models are turned into ML APIs to be available for download and use by the public.

Although our paradigm for future models offers a path forward, it is critical to recognize existing constraints. These include issues with data quality, interpretability of the models, and possible biases in training datasets. These problems must be resolved to successfully integrate models into clinical practice. We can get beyond these obstacles and raise the acceptance of AI technologies in healthcare settings by increasing the diversity of data and creating more transparent models.

6 Discussion

This study effectively analyzed the performance, features and techniques used in Deep Learning computational models for the prediction of liver cancers. Deep Learning has grown to be a tremendous help in medicine, especially in the field of disease diagnosis. These models have been shown to play a great role in helping doctors' decision-making in medical processes. The highest sensitivity achieved across the models was 100%, and 99.38% highest accuracy. However, another study showed 97% accuracy, and more studies showed accuracy and precision to be greater than 90%. Various deep learning techniques have been utilized, including CNNs, SVMs, FCNs, and other novel combinations such as the SCCNN discussed. Deep Learning models are beneficial because they meticulously search the input data for patterns, extracting features indicative of cancers within the liver. This ability of DL AI enables recognition of tumors in the liver with ease and minimal manual interaction.

Furthermore, AI has the potential to significantly decrease diagnosis time, allowing for a proactive approach to treatment methods. AI also reduces the involvement of potential viewer biases when diagnosing liver cancers non-invasively, such as through imaging [84]. This is possible because they rely purely on previously annotated and factual data and are standardized to allow for reliable diagnosis.

Before we move on, it's important to acknowledge the logistical and other difficulties associated with the clinical implementation of AI-based predictive models. The high cost of AI technologies can be a significant obstacle. Developing, deploying, and maintaining such systems usually requires substantial financial investment. Integration with existing medical infrastructure is another issue likely to come across since it is tedious to incorporate these models to ensure compatibility with current electronic health records and other clinical systems. Additionally, user acceptance is crucial: healthcare professionals may be reluctant to adopt new technologies because of various concerns about the reliability of AI-driven models and the need for extensive training to effectively utilize these tools.

Table 4 Confusion matrix

Actual condition	Positive condition	Negative condition
Positive condition	TP _n	FP _n
Negative condition	FN _n	TN _n

6.1 Research investigation 1: evaluation parameters for AI ML/DL predictive models

Evaluative parameters are the measurements used to judge the diagnostic and predictive ability of AI models. These metrics are also crucial to compare results between models and even compare to human experts to understand the real-world practicality of the model at hand. They are also used to understand the advantages and disadvantages of the models. The evaluation parameters most commonly used and those that are mentioned in this paper are described below. It is significant to analyze the confusion matrix before understanding the evaluation parameters. The confusion matrix, as shown in Table 4, plays a significant role in statistical classification. The layout of the table allows to visualize the performance of a model.

- Area under the curve (AUC): AUC refers to a measure of the ability of a model to distinguish between given classes, and is a range from 0 to 1. 0 represents exceedingly bad, whereas 1 represents very good performance. AUC curve is a measure and summary of the ROC (receiver operating characteristics) curve, which shows the true positive rate against the false positive rate for readers to compare.
- Accuracy (Acc_n): Accuracy can be characterized as the ratio of correctly identified cases by the model to total cases, as shown in Eq. (3).

$$Acc_n = \frac{TP_n + TN_n}{TP_n + TN_n + FP_n + FN_n} \quad (3)$$

- Sensitivity/Recall (R_n): Sensitivity is characterized as the ability of the model to correctly identify positive cases. Equation (4) shows this:

$$R_n = \frac{TP_n}{TP_n + FN_n} \quad (4)$$

- Precision (P_n): Precision is the ratio of correctly identified positive outcomes to total positive outcomes returned by the classifier. Mathematically, it is shown in Eq. (5).

$$P_n = \frac{TP_n}{TP_n + FP_n} \quad (5)$$

- Specificity (Sp_n): Specificity refers to the ability of the model to correctly identify negative cases. Equation (6) shows this.

$$Sp_n = \frac{TN_n}{TN_n + FP_n} \quad (6)$$

6.2 Research investigation 2: common available datasets for liver cancers

In Table 5 we have presented various publicly available datasets for liver cancers that include most of the diagnostic procedures previously discussed and additional types.

6.3 Research investigation 3: research gaps identified in current research

This section will be moving on from the challenges already discussed in Table 3. As previously discussed, the liver is a site prone to metastases in cancer patients. Because of this, the major problem that diagnostic systems aim to overcome is to be able to differentiate between PLCs and secondary malignant tumors in the liver. This problem not only troubles experienced professionals, but is a major research gap in the field of Deep Learning diagnostic models; specifically, models that utilize imaging datasets (MRI, CT, US, etc.) struggle with this. This was first observed in [67]: this study mentioned that one of the major limitations in the field is the differentiation of liver masses. In the 2D CT slices used for the study, early HCC tumors appeared similar in appearance and attenuation with dysplastic nodules in the liver. Furthermore, MRI screening also has this limitation. This study [68] used a diffusion-weighted MRI dataset and observed that DW-MRI

Table 5 Descriptions of commonly used datasets for training AI for liver cancer diagnosis

Dataset name	Dataset link	Type of dataset	Description
3D liver and liver tumor segmentation	https://www.kaggle.com/datasets/gauravdukkaiti/3d-liver-and-liver-tumor-segmentation/data	3D CT and MRI	To ensure generalizability, the dataset includes a diverse set of patients covering a range of ages, genders, and ethnicities. Each medical image has also been annotated by professionals. It includes a variety of pathological variations in tumors (size, benign, malignant, etc.). 3D data helps train for volumetric information and real-world data
Liver tumor segmentation	https://www.kaggle.com/datasets/andrewmvd/liver-tumor-segmentation/data	CT	The dataset consists of 130 CT scans of the liver and of liver lesions intended for liver tumor segmentation. The dataset was extracted from the LITS challenge. The data was provided by multiple clinical sites over the world to ensure diversity in the data
The cancer genome atlas liver hepatocellular carcinoma collection	https://www.cancerimagingarchive.net/collection/tcga-lihc/	MRI, CT and PT	The dataset is a free, public dataset consisting of data from 97 subjects, giving a total of 125,397 images. Vast quantity helps the adaptability of the model however this dataset focuses only on HCC cases so is not feasible when trying to diagnose cancers other than HCC
High-sensitive circulating microRNA panel for early detection of hepatocellular carcinoma	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE113740	MIRNA Profile	In such datasets, profiling of non-coding RNA is done using an assay for several patients. Here, serum micro-RNA profiles of 1817 samples were analyzed, with patients of HCC, non-cancer, hepatitis, and liver cirrhosis. The expression of serum MiRNA combinations can be used as a classifier for liver cancer in patients. Having profiles of a diverse set of diseases helps specify which upregulations are linked to liver cancer
Gene expression profiles of human hepatocellular carcinoma	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE45435	Gene expression profiles	This dataset consists of 31 gene expression profiles of HCC samples. The main aim was to understand the differences in gene expression of young and old HCC patients. Understanding gene expression in different diseases can help link certain genes to look out for during diagnosis
Single cell transcriptome of human intrahepatic cholangiocarcinoma	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE181878	Transcriptome sequencing	Unlike most datasets, this dataset focuses on ICC. The set involves single-cell transcriptome sequencing data of 3 samples (plasma, tumor, and paracancerous tissue) from an ICC patient. At a single-cell level, this dataset helps understand and detect unique cell heterogeneity within ICC cells, and to identify variations in cell types in different tumor environments. This dataset can be used to understand the tumor micro-environment in ICC patients to classify the particular cancer. However, only 1 patient's data is used, so its not sufficient and generalizable

shows a slower signal decay at high b-values for both primary and metastasized tumors, reducing the study's precision in distinguishing between the two. Because of the large variety of possible tumors in the liver, and the limitations that come when using imaging as the sole diagnostic tool, misdiagnosis and false positives often come about.

This limitation also reduces the independence of the models. A recurring trend was that the accuracy of the DL model-assisted doctors set was the highest, likely because the doctors are able to better utilize clinical information in their decision-making, adding to the ability of the model alone.

This problem continues between benign and malignant masses as well. A high false positive rate was common between models. Conditions such as hepatic hemangiomas, cysts, and focal nodular hyperplasia can mimic liver tumors when using imaging techniques, giving false positives. However, it is likely that the use of 3D imaging has the potential to reduce this because of the additional information it provides.

Another area of research that should be further explored is the use of 3D imaging data, including 3D CT and MRI, in combination with 3D neural networks. A vast majority of the papers found and discussed utilized imaging datasets, specifically CT, but only few utilized 3D tomographic data. Reliance on 2D data presents the models with various limitations. Firstly, 2D CNNs analyze slices of data individually, missing out on crucial volumetric information. They don't provide enough information about tumor morphology, including its interactions with surrounding organs and tissue, its shape, and location. Additionally, specifically in the liver, attempts to delineate tumors spread across multiple 2D slices often result in errors like missing out on parts of the tumor due to "multi-focal" tumor appearance [68].

Moving on, many machine learning models relied on static images as their primary datasets. This refers to simple CT, CECT, MRI, etc. The advantages of a temporal analysis through sequential imaging techniques appeared to be underutilized. Few studies implemented dynamic CT/MRI, or PET scans for training their models, even though they provide additional information. This is likely because of the increased training time and computational complexity that comes along with dynamic imaging; however, such techniques allow observations on the tumor behavior (see previous section on dynamic imaging), aiding in early diagnosis and prognosis of cancers. Utilizing this data would likely provide models with more discernible features and characteristics associated with specific cancers based on their interactions with surrounding tissue—which can be observed through dynamic imaging.

The datasets used also bring about other problems. Commonly, models are trained on data from one specific medical center, and since imaging protocols like contrast phases, resolution, etc. are not always standardized across centers, models end up doing worse in the validation set, or in real-world applications. This is a big reason behind the delay in FDA approval for the clinical use of many AI models. It is vital that in training, models are trained on a variety of datasets from different centers, that also take into account a diverse group of patients and diseases, in this case: benign masses, HCC, ICC, secondary liver tumors, etc. Overfitting is also prevalent and similar to what was described above. Overfitting occurs when a model gets too acquainted with the training data and performs extremely well on the training data in comparison to the testing data or validation set. This occurs when enough training data isn't provided (the data set is too small), where models get too familiar with the noise and unimportant details in the training data and are unable to identify a general pattern.

A problem that was observed when using a miRNA dataset was the lack of current literature on the topic. The paper analyzed [69] identified multiple relevant miRNA biomarkers associated with liver cancers. However, because of the little research that exists on some miRNAs, the study was unable to achieve its potential maximum accuracy. This is because some important data values were missing. Although miRNA profiling serves as a great means to differentiate and diagnose tumors and various diseases, lacking literature creates a strong limitation for using such datasets.

A lack of existing literature also impacted the paper analyzed using a DNA methylation dataset [70]. Although the study presented with a high accuracy with novel methylation sites associated with HCC, longitudinal studies are required to accurately confirm the prognostic value of these markers and their link to liver cancer. Lastly, the study analyzed relied on the use of tumor tissue biopsies to obtain methylation profiles from patients, causing unwanted trauma for diagnosis. Although newer research has found alternative solutions to this, this paper still was considered unnecessarily invasive, which challenged its diagnostic efficacy.

6.4 Research investigation 4: suggestions for future research

Certain areas and algorithms were observed to show notable performance and efficacy improvements within the models. These are discussed below.

- The long-lasting problem in the space of liver cancer diagnosis is between PLCs and SLCs. For this, it is vital to consider the importance of a structured AI model, incorporating model imaging and miRNA tools. For a first diagnosis, imaging is likely the right approach as it has a larger area of view and is easier to interpret. However, when attempting to differentiate PLCs and SLCs, and when biopsies, or now peripheral blood sampling (utilization of ctDNA) are doable for miRNA profiling, evidence suggests that these may also be a wise approach. miRNA play a crucial role in gene expression and often show aberrant patterns in cancer tissue. miRNA profiling can significantly enhance the early detection and differentiation of primary versus secondary liver tumors. This is because genetic changes are often more reliable and clearly evident than imaging alone. Future models could incorporate miRNA expression data to complement imaging data, leading to easier and more efficient early detection. The same applies to a broad range of genomic-based models, including but not limited to DNA methylation, gene signatures and proteomics. Furthermore, multi-omics have shown to often be superior to the single-omics models discussed above [85]. showed through their research that multi-omics models (trained on 3-omic data) showed greater prediction than single-omics-based DL models. The authors of the study claimed and showed that multi-omics data is better for model training in comparison to single-omics; however, it is important to acknowledge the difficulty of real-world implementation of multi-omics data and the difficulty involved in conducting multiple sequencing technologies, attaining patient consent while avoiding patient stress/risk, and collecting data overall. Also, multi-omics data helps in overcoming the fact that medical data such as tumor data has high intra-tumoral heterogeneity, where within a single tumor there can be significant genetic, epigenetic, and phenotypic diversity. Utilizing only one time of data creates limitations as such models may not capture the full complexity of the tumor.
- When moving towards clinical trials for a deep learning model, it is important to prepare in advance for a broader testing cohort. Because of this, it is vital to consider rarer cancers, which are often not available in online datasets. A possible solution to this is to utilize GANs (generative adversarial networks)—these networks are able to generate synthetic medical images that can then potentially address the issue of a small or lacking dataset [86]. Medical data helps these networks augment existing datasets and thus improving their performance on underrepresented tumor types.
- Deep learning image reconstruction (DLIR), a technique utilized by [81, 82] has shown significant implications in the differentiation between metastatic and primary cancers in the liver. DLIR are a series of deep learning algorithms used to reduce noise and artifacts and enhance resolution and contrast from imaging modalities like PET and CT scans. DLIR results in clearer and more detailed medical images. Along with aiding diagnostic precision, it also allows for better visualization of small lesions that may not be discernible using traditional image reconstruction/enhancement methods [87]. Moreover, DLIR facilitates more accurate delineation of tumor boundaries, which is critical in diagnosis and prognostication of this cancer. The enhancements done through DLIR help radiologists distinguish between different tissue types and identify abnormalities with greater accuracy—as shown in [82]. DLIR also produces high quality and noise reduces images with relatively lower radiation doses, thereby minimizing patient exposure and allowing for frequent monitoring [88]. Lastly, the clinical implementation of DLIR systems is also relatively more feasible. This is largely because unlike most, DLIR algorithms are not fully autonomous; instead, they serve as diagnostic aids to radiologists.
- The usage of curated supervised attention mechanisms is also suggested. A supervised attention mechanism (SAM) in DL refers to a method of guiding a model's focus on relevant parts of input data based on external annotations done by expert radiologists/doctors. This has been shown to enhance AI models' ability to prioritize relevant features, significantly improving accuracy [80]. In medical screenings, SAMs allow radiologists to annotate on critical regions such as tumors or lesions and teach the model to better differentiate through their notes. Schmauch et al. [80] used a SAM and thereby showed improvement in the model's ability to differentiate between 5 categories, including HCC and SLCs. Radiologists manually identified abnormalities and used bounding box annotations when training the model. This aided in classification as mentioned before, but also improved the interpretability of the model. By overlaying attention maps on the ultrasound images, the researchers could visualize which areas the model focused on most, allowing them to understand where the model made errors and where it succeeded—essentially supervising the model's diagnosis. SAMs showed in the studies [72, 80] that they have potential to increase accuracy when model's are required to classify inputs into non-binary classes (3 or more categories, for example: HCC, ICC, benign, SLC).
- Coming to preprocessing techniques used by the papers analyzed, the combination of Watershed and Gaussian Mixture Model (GMM) stands out by far for imaging datasets. This model showed a 100% sensitivity and 99.38% accuracy. Watershed transform aids in image segmentation by treating the grayscale image as a topographic surface representing intensities as peaks, hills and valleys. This tool is used to delineate regions, specifically the liver and tumor regions.

By doing this preprocessing, the load and data provided onto the classifier are reduced and the region of interest is isolated, therefore likely resulting in a higher accuracy. The GM model is used to cluster the pixels in an input image into different groups based on their intensity values, essentially clearly differentiating tumor and non-tumor regions. This is further used to isolate the ROI and has been shown to increase both accuracy and sensitivity. Together, these models show a high potential for clinical use. It is important to note the added computational complexity that results from these add-ons, however. The main purpose of this is to emphasize the significance of preprocessing and segmentation algorithms in models, not just referring to GMM and Watershed. However, the immensely high accuracy must be critically evaluated and not taken at face value. These results were achieved with a relatively small dataset of 225 CT images, which may not be sufficient for robust conclusions. The study used a threefold cross-validation method rather than the more robust 5 or tenfold. Additionally, the high performances such as this suggest potential for overfitting. Future studies should validate these findings on larger, more diverse datasets with robust cross-validation methods to ensure the results are generalizable and reliable in clinical practice.

- Sparsity-based classification may be a solution to the above-mentioned problem of computational complexity. To address the potential for regular clinical use, it is necessary to maintain computational complexity and user interpretability. This method cuts down on the features being analyzed by identifying and focussing on the most significant ones. It also has shown promise in its use to reduce false positive rates and increase model efficiency.
- Siamese networks use a twin mechanism, where two identical neural networks are trained in parallel on two different inputs. This mechanism is highly useful in differentiating objects of similar types and therefore may be worth researching in its implications in differentiating between PLCs and between PLCs and SLCs.
- Alongside the previously listed advantages of 3D data and model usage, the 3D architecture proposed in the paper [68] also removes the need for any preprocessing steps. This refers to cropping, ROI annotating, etc. Although 3D models and data increase the load put on the model, the removal of such add-ons reduces the complexity in an attempt to increase efficacy.
- Dynamic and Contrast-enhanced imaging showed higher accuracies in the analyzed studies, likely due to the added data with dynamic images and easier delineation and segmentation through contrast.
- As mentioned in the previous section, dataset handling is also vital to create generalizable models that are practical to use in real-world scenarios. Medical datasets commonly consist of class imbalance, where different instances of each class are not equally distributed. GANs mentioned previously are one approach to this problem; however, we also emphasize the importance of oversampling and undersampling techniques for class balancing. This has been elaborated in Sect. 5.
- Lastly, we also want to emphasize the astonishing growth and implementation of transfer learning in the field of medical images. As detailed thoroughly in Sect. 3.3, transfer learning modes aid in reducing manual interaction, delineation, segmentation, and can be converted into hybrid models enabling them with higher precision and accuracies. We encourage further investigations into individual and hybrid transfer learning models.

We propose to evaluate the performance of the proposed machine learning approach on multiple datasets representing varied patient demographics. Also, we acknowledge the need for further validation in real-world clinical environments for assessing the generalizability. In the future, we aim to address all the issues raised in the study regarding previous research and focus on testing the model with real-world datasets in distinct clinical settings. This can assure the model's generalizability, thus enhancing its clinical utility and reliability [89, 90].

6.4.1 Clinical impacts of AI-based predictive models

- IBM Watson oncology: Oncologists can use IBM Watson for Oncology to make better treatment decisions. Watson helps hospitals globally, like as Manipal Hospitals in India, personalize cancer treatment programs by examining a patient's medical history, genetics, and the latest research [91].
- Google AI for diabetic retinopathy: Google AI's deep learning algorithm can detect diabetic retinopathy, which can cause blindness if untreated. Indian clinics use this technique to swiftly and precisely determine at-risk patients. This AI solution prioritizes urgent patients, decreasing specialist burden and assuring prompt interventions [92].
- Aidoc for radiology: Aidoc helps radiologists discover medical picture anomalies with AI. It helps radiologists prioritize critical cases like CT scan strokes and cerebral hemorrhages [93].

- PathAI for pathology: Machine learning tools from PathAI help pathologists diagnose diseases from tissue samples. It increases pathology results consistency and reduces diagnosis errors by giving pathologists AI-powered insights [94].

7 Conclusion

Liver cancer is a highly prevalent form of cancer, with nearly 800,000 new cases annually, and this number is expected to increase over time. However, studies have shown that artificial intelligence has made significant progress in tackling this issue by creating predictive DL models that have shown good accuracy and performance for diagnosing liver cancers. Despite these advancements, our research identifies several gaps that need to be addressed to enhance the efficacy of predictive models. A critical weakness identified across many studies is the inability of AI-based models to efficiently differentiate between PLCs and SLCs. Additionally, underutilization of 3D datasets, problems with dataset origin, dataset type, and more have also been identified as areas that need to be addressed in future research. This investigative review answered key questions and the results are summed as follows: We detail the evaluation parameters that should ideally be used to evaluate performance of future predictive models. We also explain common liver cancer datasets, detailing various types, and provide a comprehensive list of research gaps in current literature. Lastly, we have also listed potential suggestions for future research, backed by evidence from our research. This review can guide the development of future models by providing crucial information about valuable algorithms, preprocessing techniques, and datasets that can be used to achieve high performance in AI-based models in the future. These insights, along with previous research, will help AI models improve as efficient decision-support systems for doctors in the field of cancer diagnosis.

Author contributions S. Grover: Conceptualization, Writing – original draft, Data Collection, Critical Review S. Gupta: Writing – review & editing, Supervising.

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Data availability The datasets presented in this study can be found in online repositories. The names of the repository/repositories and accession number(s) can be found in the article.

Declarations

Competing interests The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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